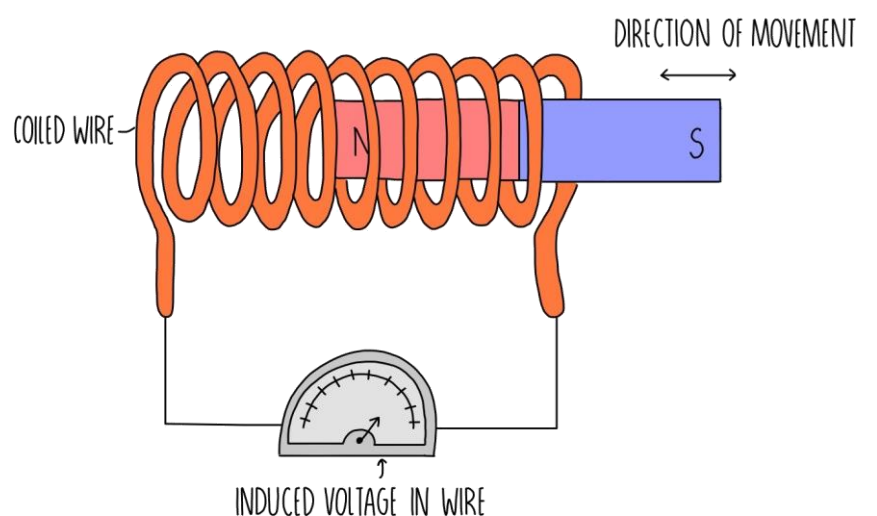
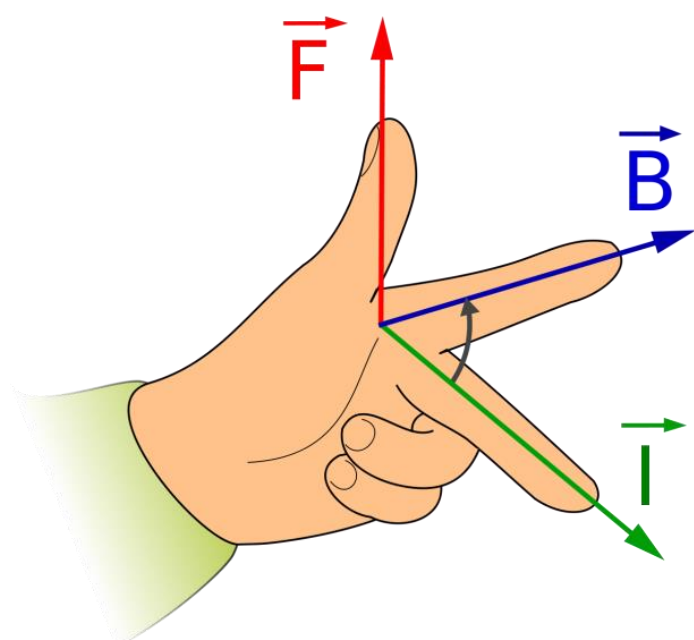
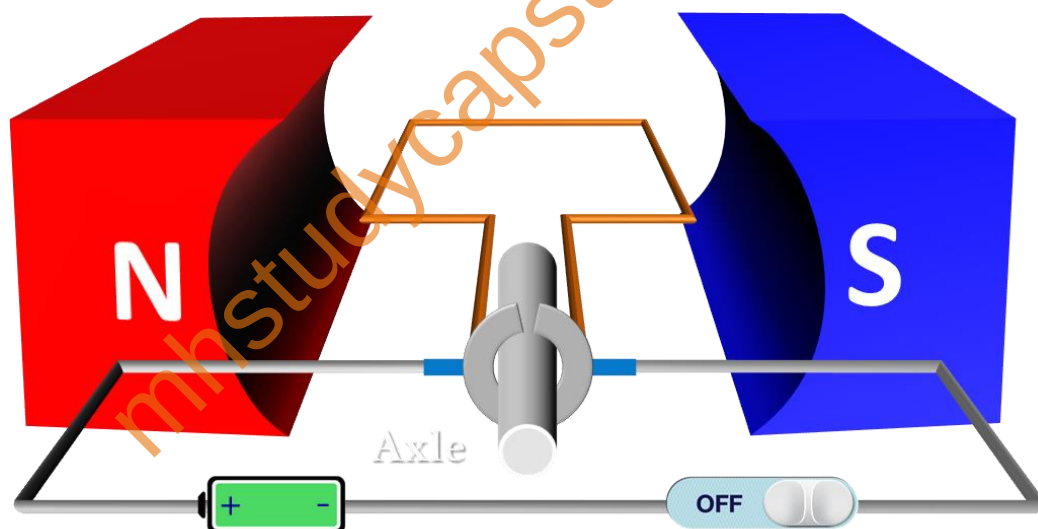


Chapter 4

Effects of Electric Current



1. Tell the odd one out. Give proper explanation.

a. Fuse wire, bad conductor, rubber gloves, generator.

The odd one out is generator. It is an electrical device for producing electricity. Fuse wire, bad conductor and rubber gloves have high resistance and are used for blocking electricity. Thus, they can be used as a safety measure against heavy electricity.

b. Voltmeter, Ammeter, galvanometer, thermometer.

The odd one out is thermometer. It is an instrument for measuring the temperature of a body. Rest of the three are electrical instruments based on the phenomenon of electromagnetism and are used for measuring some electrical parameters such as current and voltage.

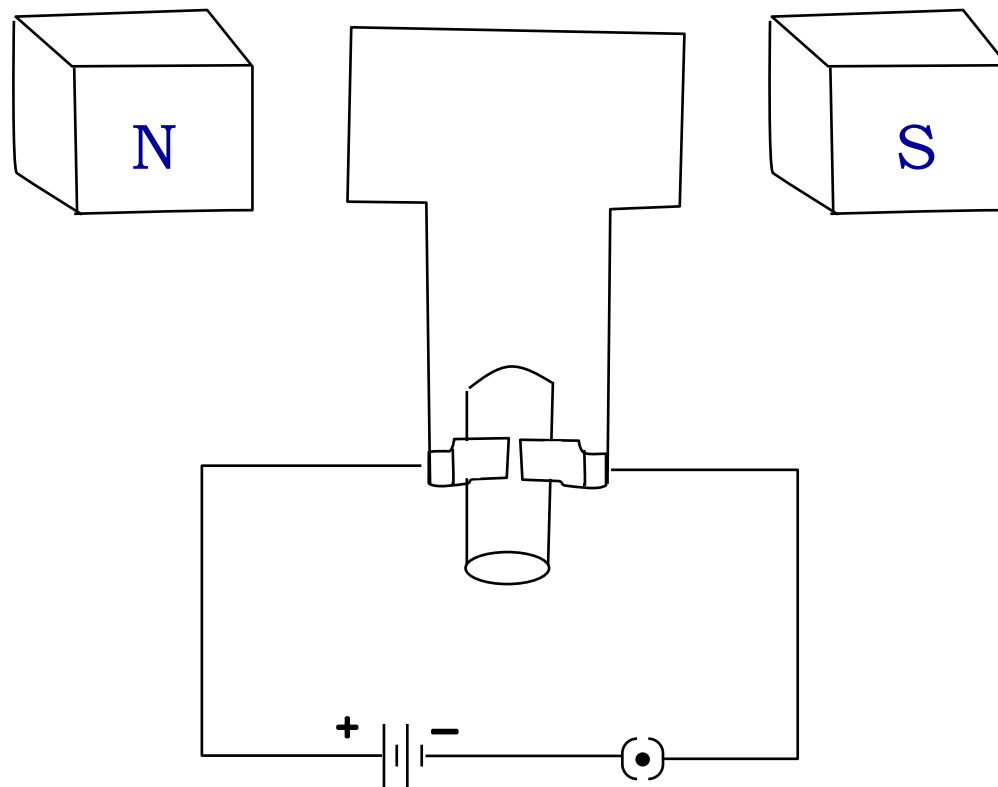
c. Loudspeaker, microphone, electric motor, magnet.

The odd one out is magnet. Loudspeaker, microphone and electric motor are based on the phenomenon of electromagnetism.

2. Explain the construction and working of the following. Draw a neat diagram and label it.

a. Electric motor

P.T.O



Electric motor: A device which converts electrical energy into mechanical energy.

Construction: Electric motor consists of following parts:

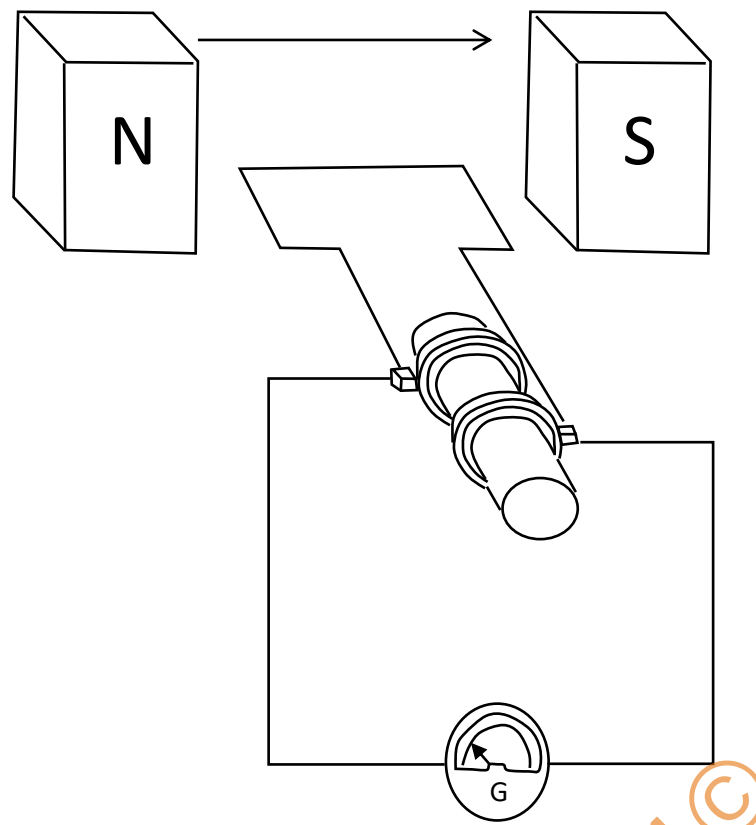
- i. **Rectangular coil:** Rectangular loop of copper wire having resistive coating.
- ii. **Strong magnet:** Coil is placed between the north pole and south pole of a magnet (such as a horseshoe magnet) in such a way that its branches AB and CD are perpendicular to the direction of magnetic field.
- iii. **Split ring:** The two ends of the loop are connected to the two halves (X and Y) of the split ring.
- iv. **Axle:** The two halves of the ring have resistive coating on their inner surfaces and are tightly fitted on the axle.

- v. **Carbon brushes:** The two halves of the split ring, X and Y, have their outer conducting surfaces in contact with the two stationary carbon brushes, (E and F), respectively.

Working:

- i. When the circuit is completed, the current flows in the branch AB of the loop from A to B through the carbon brushes E and F.
- ii. Since the direction of the magnetic field is from north pole to south pole, according to the Fleming's left-hand rule, a force is exerted on the branch AB in downward direction and CD in upward direction.
- iii. Thus, the loop and the axle start rotating in an anticlockwise direction.
- iv. After half rotation, the current in a loop starts flowing in the direction DCBA.
- v. Therefore, a force is exerted on the branch CD in downward direction and on the branch AB in the upward direction, and the loop continues to rotate in the anticlockwise direction.
- vi. Thus, the current in the loop is reversed after each half rotation and the loop and the axle continue to rotate in the anticlockwise direction.

b. Electric Generator (AC)



A device which converts mechanical energy into electrical energy in the form of alternating current is called AC Generator.

Principle of working:

Electric generators work on the principle of electromagnetic induction.

Construction:

- i. **Rectangular Coil:** Copper wire coil ABCD.
- ii. **Strong magnets:** Coil is kept between the two pole pieces of a magnet.
- iii. **Conducting Rings:** The two ends of the coil are connected to the conducting rings R_1 and R_2 .
- iv. **Axle:** Both the rings are fixed to the axle, but there is a resistive coating in between the ring and the axle.
- v. **Carbon brushes:** The stationary carbon brushes B_1 and B_2 are connected to a galvanometer.

Working:

- i. When the armature coil ABCD is rotated by external force in the magnetic field provided by strong magnets, it cuts the magnetic lines of forces. Thus, the changing magnetic field induces current in the coil.
- ii. After rotation of axle, arm AB moves upwards, while arm CD moves downwards. Therefore, ABCD rotates in the clockwise direction.
- iii. As per the Fleming's right-hand rule, the current is induced in the Coil and flows from $A \rightarrow B$ and $C \rightarrow D$. Current flows in the direction ABCD.
- iv. Current flows from B_2 to B_1 in the first half of the revolution in the external circuit.
- v. After half revolution, arms AB and CD exchange their positions. Due to this, the induced current flows in the direction DCBA. But arm AB is in contact with B_1 , through slip ring and arm CD is in contact with B_2 .
- vi. In the second half cycle, in the external circuit, electric current flows from B_1 to B_2 , i.e., opposite to the previous half rotation.
- vii. This repeats after every half rotation, and alternating current is produced.

3. Electromagnetic induction means-

- a. Charging of an electric conductor.
- b. Production of magnetic field due to a current flowing through a coil.
- c. Generation of a current in a coil due to relative motion between the coil and the magnet.
- d. Motion of the coil around the axle in an electric motor.

4. Explain the difference :

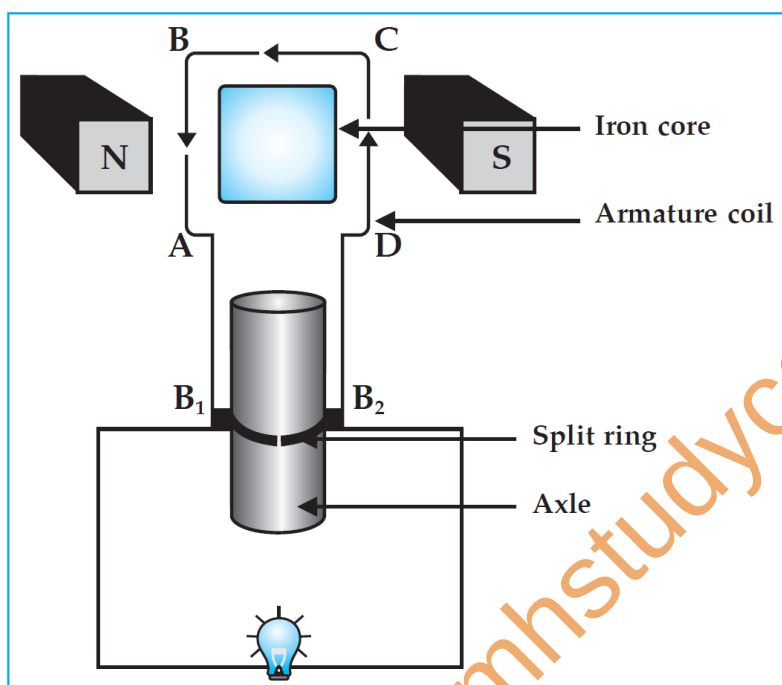
AC generator and DC generator.

AC Generator	DC Generator
AC generator is a mechanical device that converts mechanical energy into AC electrical power.	DC generator is a mechanical device that converts mechanical energy into DC electrical power.
In an AC generator, the current produced reverses direction periodically.	In a DC generator, the current produced flows only in one direction.
AC generators have slip-rings.	DC generators have commutators.
In an AC generator, the output current can be either induced in the stator or in the rotor.	In a DC generator, the output current can only be induced in the rotor.

5. Which device is used to produce electricity? Describe with a neat diagram.

- a. Electric motor
- b. Galvanometer
- c. Electric Generator (DC)
- d. Voltmeter

Ans: The device that converts mechanical energy into the electrical energy in the form of direct current is called DC generator.



Principle of working:

DC generator works on the principle of electromagnetic induction.

Construction: Electric motor consists of following parts:

- i. **Rectangular coil:** Rectangular loop of copper wire having resistive coating.
- ii. **Strong magnet:** Coil is placed between the north pole and south pole of a magnet (such as a horseshoe magnet) in such a way that its branches AB and CD are perpendicular to the direction of magnetic field.

- iii. **Split ring:** The two ends of the loop are connected to the two halves (X and Y) of the split ring.
- iv. **Axle:** The two halves of the ring have resistive coating on their inner surfaces and are tightly fitted on the axle.
- v. **Carbon brushes:** The two halves of the split ring, X and Y, have their outer conducting surfaces in contact with the two stationary carbon brushes, (E and F), respectively.
- vi. **Electric bulb or Ammeter :**
 - a) The output is shown by the glowing bulb connected across the carbon brushes.
 - b) Ammeter is connected in the circuit to detect the presence of electric current in the circuit.

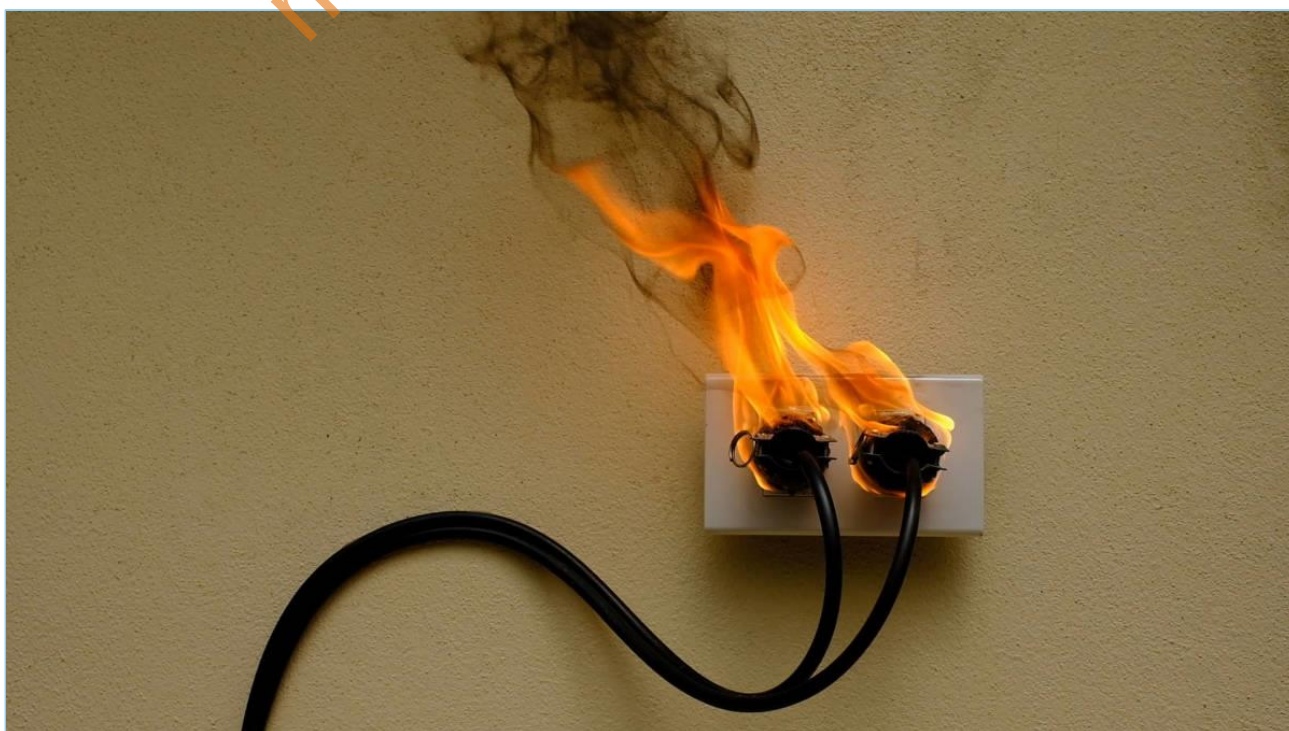
Working:

- i. The axle is rotated with a machine from outside.
- ii. When the armature coil of the generator rotates in the magnetic field, electric potential difference is produced in the coil due to electromagnetic induction.
- iii. This produces a current as shown by the glowing of the bulb or by a galvanometer.
- iv. The direction of the current depends on the sense of rotation of the coil.

- iv. In a DC generator, one brush is always in contact with the arm of the coil moving up while the other brush is in contact with the arm of the coil moving down in the magnetic field.
- v. Hence, the flow of the current in the circuit is always in the same direction and the current flows so long as the coil continues to rotate in the magnetic field.

6. How does the short circuit form? What is its effect?

- i. If a live wire (phase wire) and a neutral wire come in direct contact or touch each other, short-circuiting takes place.
- ii. Due to a fault in the equipment or if the plastic coating on the 'live' and the 'neutral' wires gives way, the two wires come in contact with each other and a large current flows through it producing heat.
- iii. If any inflammable material (such as wood, cloth, plastic etc.) exists around that place it can catch fire.



7. Give Scientific reasons.

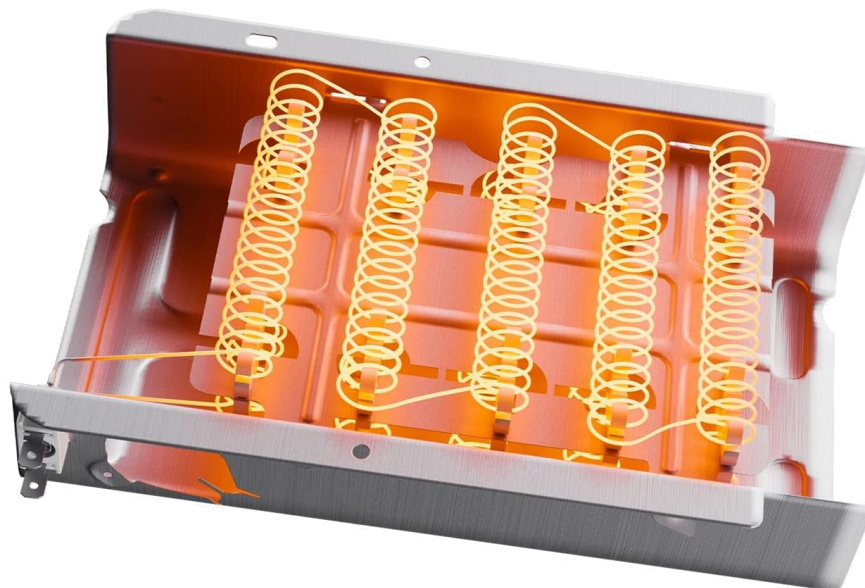
a. Tungsten metal is used to make a solenoid type coil in an electric bulb.

- i. The electric bulb works on the principle of heating effect of electric current.
- ii. The intensity of light emitted by the filament of an electric bulb depends on the temperature of the filament.
- iii. When electric current is passed through the filament, a large amount of heat is generated and filament becomes hot (nearly 3400°C), then emits light.
- iv. The melting point of Tungsten is very high, so that filament can be heated to a high temperature without melting.
- v. Hence, the filament of electric bulb is made up of Tungsten.

b. In the electric equipment producing heat e.g., iron, electric heater, boiler, toaster etc. an alloy such as Nichrome is used, not pure metals.

- i. The appliances like iron, electric toaster etc. works on the principle of heating effect of an electric current.
- ii. At high temperature pure metal can get oxidised and also pure metals have low resistivity.
- iii. An alloy such as Nichrome, has high resistivity and thus can be heated to a high temperature. It also does not undergo oxidation.

- iv. Thus, Nichrome is preferred over other pure metals and are used for making coil for devices working on heating effect of electric current.



c. For electric power transmission, copper or aluminium wire is used.

- i. Copper and Aluminium are good conductors of electricity.
- ii. Copper and aluminium have large number of free electrons and are highly ductile metals.
- iii. They have less resistivity, due to which these metals have high conductivity.
- iv. Hence, connecting wires in the electric circuit are generally made of copper and aluminium.



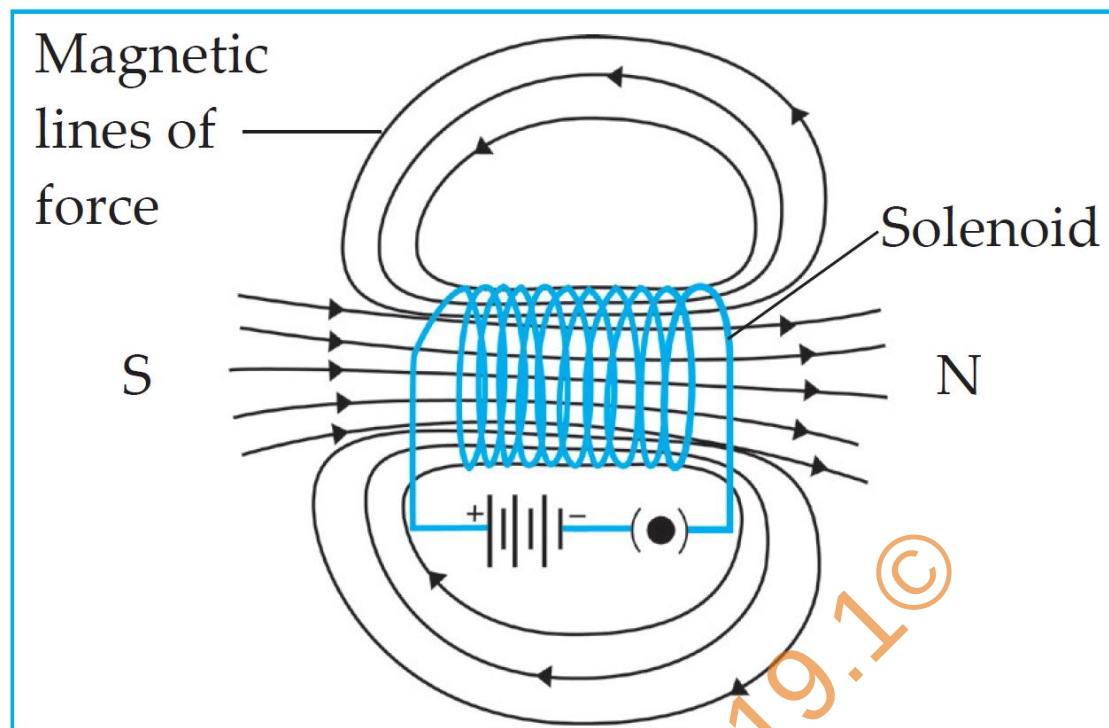
- d. In practice the unit kWh is used for the measurement of electrical energy, rather than joule.
- i. The unit of electric power 1W is a very small unit, hence 1000 W or 1 kW is used as a unit to measure electric power, in practice.
- ii. If 1 kW power is used for 1 hour, it will mean $1\text{kW} \times 1 \text{ hr}$ of electrical energy is used.
- iii. $1\text{kWh} = 1 \text{ kilowatt hour} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ Ws} = 3.6 \times 10^6 \text{ J}$.
- iv. joule is the SI unit for energy which is very small, hence on commercial scale kWh is used to measure electrical energy.

8. Which of the statement given below correctly describes the magnetic field near a long, straight current carrying conductor?

- a. The magnetic lines of force are in a plane, perpendicular to the conductor in the form of straight lines.
- b. The magnetic lines of force are parallel to the conductor on all the sides of conductor.
- c. The magnetic lines of force are perpendicular to the conductor going radially outward.
- d. The magnetic lines of force are in concentric circles with the wire as the center, in a plane perpendicular to the conductor.



9. What is a solenoid? Compare the magnetic field produced by a solenoid with the magnetic field of a bar magnet. Draw neat figures and name various components.



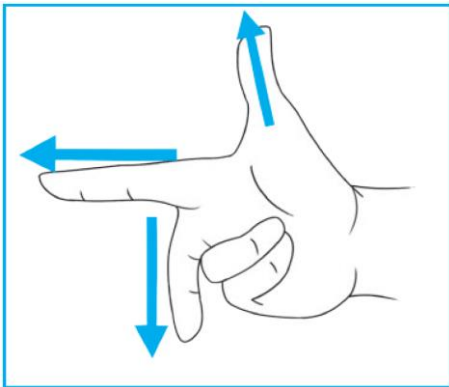
- i. When a copper wire with a resistive coating is wound in a chain of loops (like a spring) it is called solenoid.
- ii. When an electric current is passed through a solenoid, magnetic lines of force are produced in a pattern as shown in the diagram.
- iii. The properties of the magnetic field of a solenoid are very similar to the magnetic field produced by a bar magnet.
- iv. One end of the coil acts as the south pole, while the other end as the north pole.
- v. The magnetic lines of force inside the solenoid are parallel to each other.
- vi. This means that the intensity of the magnetic field within the solenoid is uniform everywhere, i.e., the magnetic field in a solenoid is uniform.

vii. Various components are:

- a. Copper wire with Resistive coating,
- b. Battery
- c. Plug key

10. Name the following diagrams and explain the concept behind them.

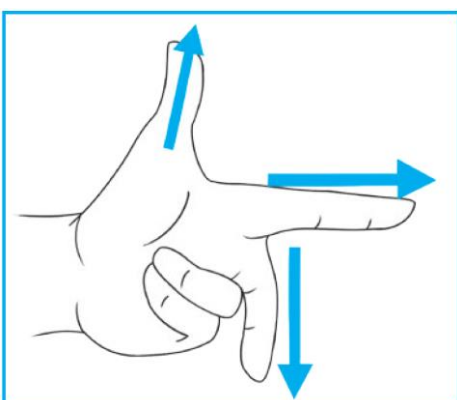
(a)



- i. Fleming's right-hand rule
- ii. This rule is used to find the direction of the induced current.

iii. Stretch the thumb, the index finger and the middle finger of the right hand in such a way that they are perpendicular to each other. In this position, the thumb indicates the direction of motion of the conductor, the index finger indicates the direction of the magnetic field, and the middle finger shows the direction of the induced current. This rule is known as Fleming's right-hand rule.

(b)



- i. Fleming's left-hand Rule.
- ii. This rule is used to find the direction of the force on the conductor.

- iii. According to this rule, the left-hand thumb, index finger, and the middle finger are stretched such that they are perpendicular to each other. If the index finger is in the direction of the magnetic field, and the middle finger points in the direction of the current, then the direction of the thumb is the direction of the force on the conductor.

11. Identify the figures and explain their use.

(a)



Figure (a) represents a fuse. An electric fuse is a safety device that protects the wiring against excessive heating caused by an excess supply of current. It melts when heavy current flows through the circuit, thereby causing the circuit to become open.

(b)



Figure (b) represents an MCB. An MCB is a device which functions as a fuse but does not require replacement. MCB trips down to break the circuit when heavy amount of current flows through it. Once the fault is rectified, the MCB is reset.

(c)

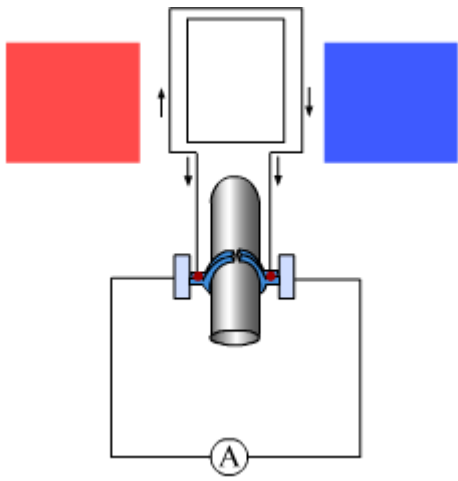


Figure (c) represents a DC generator. It is a device that generates electricity by rotating its coil in a magnetic field. Thus, it converts mechanical energy into DC electrical energy.



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INTEXT QUESTIONS

1. How do we decide that a given material is a good conductor of electricity or is an insulator?

A material which has very low electrical resistance is called a good conductor of electricity.

Examples: silver, copper, aluminium.

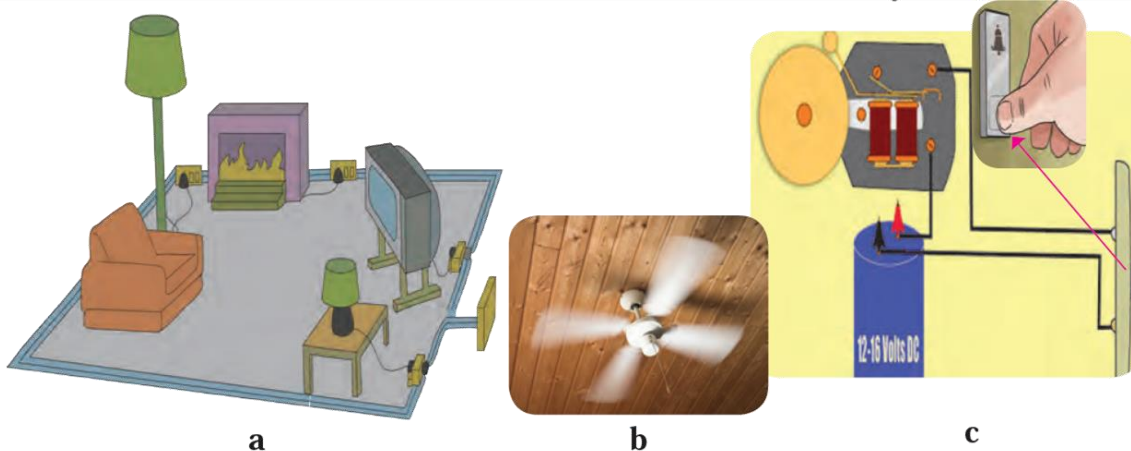
A material which has extremely high electrical resistance is called an insulator of electricity.

Examples: rubber, wood, glass.

2. Iron is a conductor of electricity, but when we pick up a piece of iron resting on the ground, why don't we get electric shock?

In a piece of iron, the net flow of electrons is zero and hence current is zero. Hence, we don't get electric shock. If we want an electric current to flow through it, then we need to connect it to a source of potential difference i.e., cell.

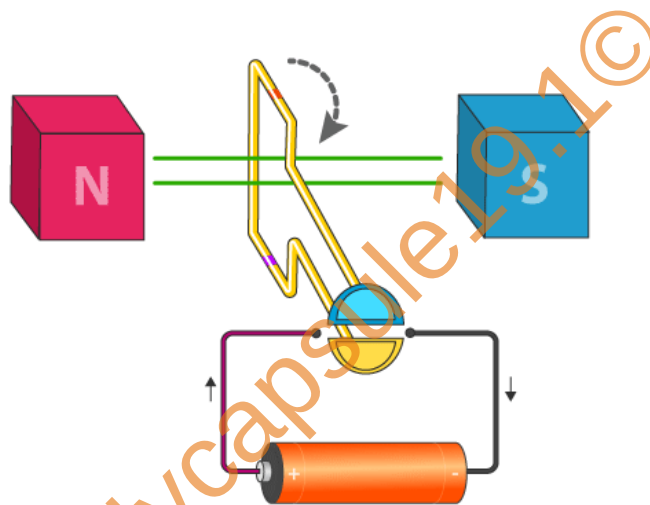
3. What do you observe in the following pictures? Which effects of electric current do you find?



- a. T.V., electric fireplace and lamps are working on heating effect of electric current.
- b. Fan is rotating due to magnetic effect of electric current.
- c. Electric bell is ringing due to magnetic effect electric current.

4. If in the circuit, the resistor is replaced by a motor, in which form will the energy given by the cell get transformed into?

If the resistor is replaced by a motor, then the energy from the cell will get transformed into mechanical energy.



5. How can we write mechanical power in a manner similar to the electrical power?

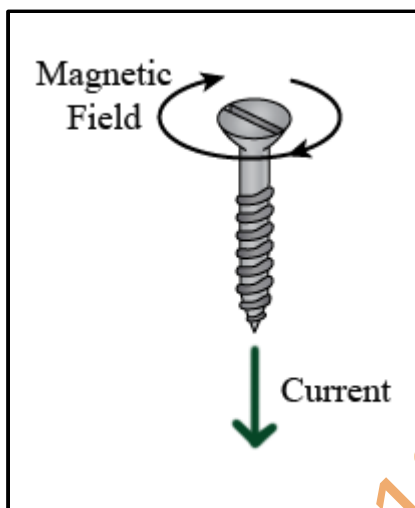
$$\begin{aligned} \text{Electrical Power} &= \frac{\text{Electric Work}}{\text{Time}} \\ &= \frac{\text{Energy consumed}}{\text{Time}} \end{aligned}$$

$$\text{Mechanical Power} = \frac{\text{Work}}{\text{Time}}$$

6. The right-hand thumb rule is called Maxwell's cork-screw rule.

What is the cork-screw rule?

According to Maxwell's corkscrew rule: Imagine driving a corkscrew in the direction of current, then the direction in which we turn its handle is the direction of the magnetic field.



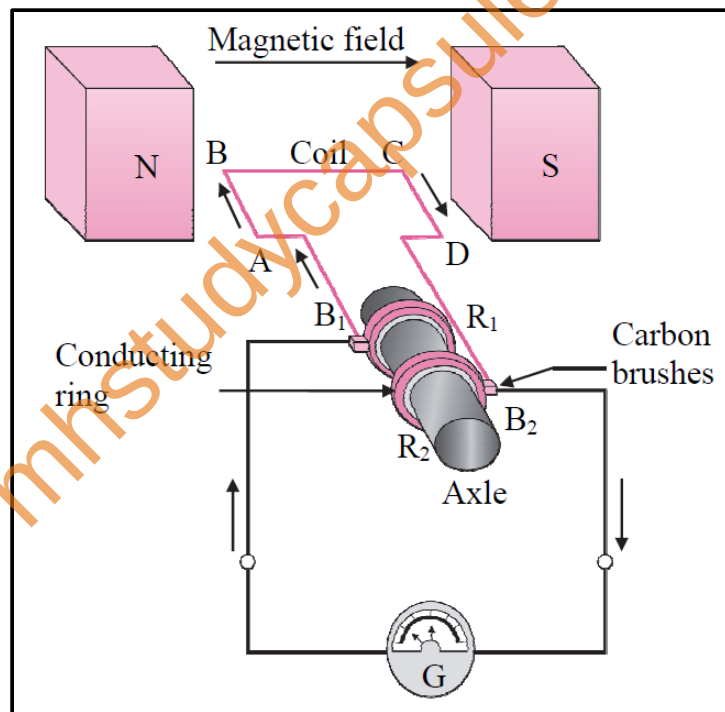
7. Why are carbon brushes used? How do these work?

- i. Brushes make contact with the split ring and are responsible for the make and break of the circuit.
- ii. As the brushes are made up of carbon (Graphite) they offer less resistance and act as a good conductor and are also cheaper to replace.

8. Draw the diagram of a DC generator. Then explain as to how the DC current is obtained. - Exercise Question 5

Previous Year Questions

1. Two tungsten bulbs of power 50 W and 60 W work on 220 V potential difference. If they are connected in parallel, how much current will flow in the main conductor? – **Numericals at the end**
2. In the electric equipment producing heat e.g. iron, electric heater, boiler, toaster etc., an alloy such as Nichrome is used, not pure metals. – **Exercise Question 7 - b**
3. Observe the following diagram and write the answers of the given sub-questions:



- a. Which instrument the above figure shows?

The above figure shows an Electric Generator (AC)

- b. Which rule is used to determine the direction of the current produced?

Fleming's Right-hand rule.

c. State the rule. – **Exercise Question 10 - a**

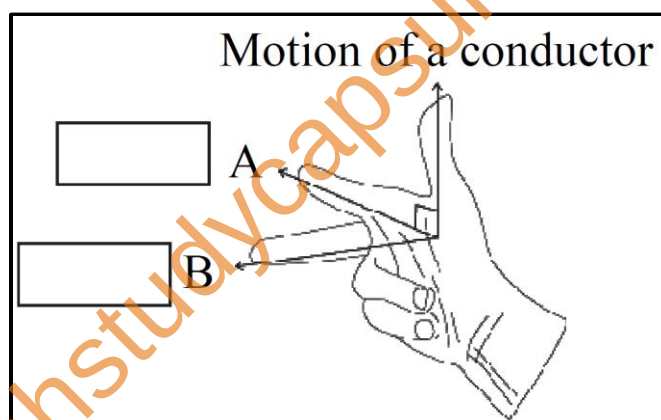
d. In which direction (B_1 to B_2 or B_2 to B_1) will the current flow in the external circuit in that situation?

The current in the external circuit is flowing from B_2 to B_1 .

e. What change will have to be made in the coil for increasing the current several times without changing the magnet?

Current can be increased several times by increasing the number of turns in the coil.

4. Observe the given figure of Fleming's Right Hand Rule and write the labels of A and B correctly.



A – Direction of magnetic field

B – Direction of induced current

5. Read the given passage and answer the following questions:

The home electrical connection consists of 'live', 'neutral' and 'earth' wires. The 'live' and the 'neutral' wires have potential difference of 220 V. The 'earth' is connected to ground. Due to a fault in the equipment or if the plastic coating on the 'live' and the 'neutral' wires gives a way the two wires come in contact with each other and a large current flows through it producing heat. If any inflammable material (such as wood, cloth, plastic, etc.) exists around that place it can catch fire. Therefore, a fuse wire is used as a precautionary measure.

a. Name the two wires having potential difference of 220 V.

The 'live' and the 'neutral' wires have potential difference of 220 V.

b. What is short circuit?

Due to a fault in the equipment or if the plastic coating on the 'live' and the 'neutral' wires gives a way the two wires come in contact with each other and a large current flows through it producing heat. This is called a short-circuit.

c. Write the function of a fuse.

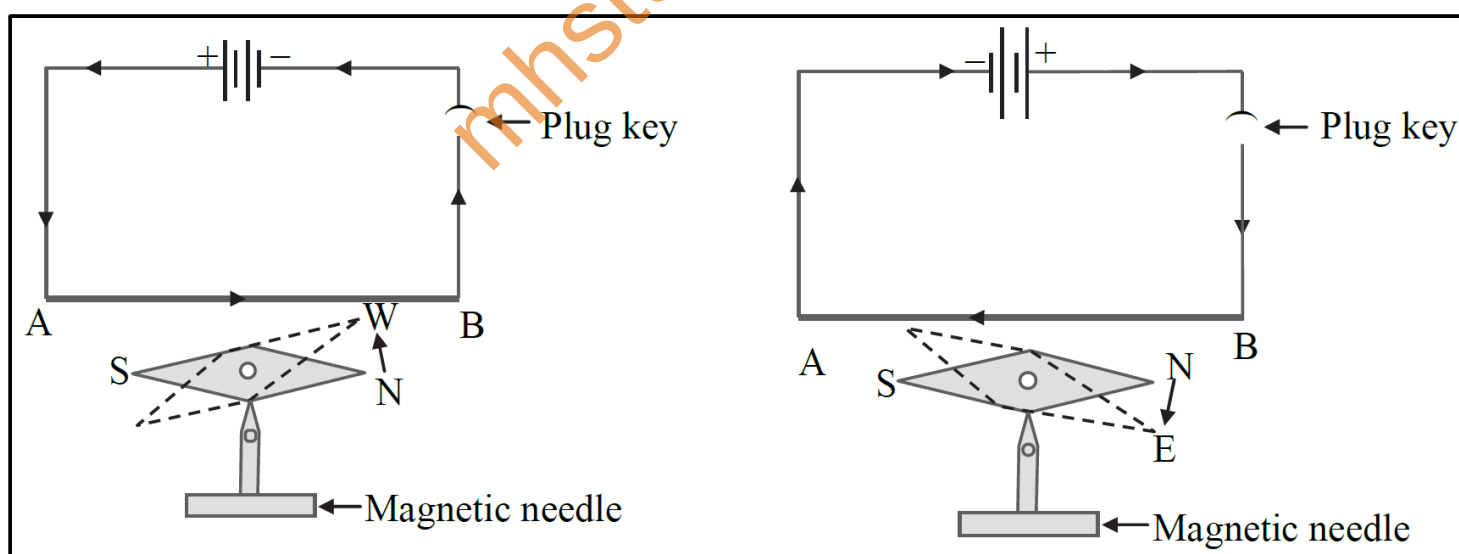
Fuse wire protects circuits and appliances by stopping the flow of any excess electric current.



6. Distinguish between 'alternating current' and 'direct current'.

Alternating Current	Direct Current
Alternating current varies in magnitude continuously. i.e., oscillatory.	Magnitude of direct current from a source remains same. i.e., non-oscillatory.
Produced by A.C. generator.	Produced by D.C. generator
This type of current is used in electrical household appliances such as electric heater, refrigerator, electric iron, etc.	This type of current is not used much for household purposes but used in batteries, torches etc.
Frequency of A.C. in India is 50 Hz.	Frequency of D.C. is zero.

7. Observe the diagrams and answer the questions:



a. Which effect of electric current is shown in the above figure?

Magnetic effect of electric current is shown in the figure

b. What will happen if the number of electric cells is increased on the magnetic needle?

Increase in the number of cells will further increase the deflection of the magnetic needle.

c. If the distance between the conductor and magnetic needle is increased, what will be the effect on intensity of magnetic field?

The strength of the magnetic field is inversely proportional to the distance from the current carrying conductor. That is, as the distance increases the magnetic field decreases.

d. If the ends of electric cell are interchanged, what will be the effect of the magnetic needle?

The deflection of the magnetic needle will be in opposite direction.

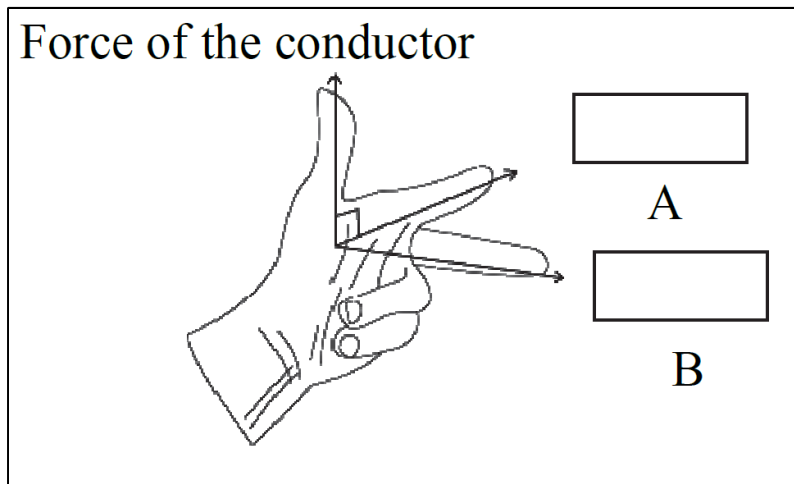
e. Write the names of any two instruments which work on magnetic effect of electric current.

Electric bell, loudspeaker

8. It is necessary to connect earth wires in home electric connection.

- i. The metallic body of electric appliances is connected to the earth by means of earth wire so that any leakage of electric current is transferred to the ground.
- ii. This prevents any electric shock to the user. That is why earthing of the electrical appliances is necessary.

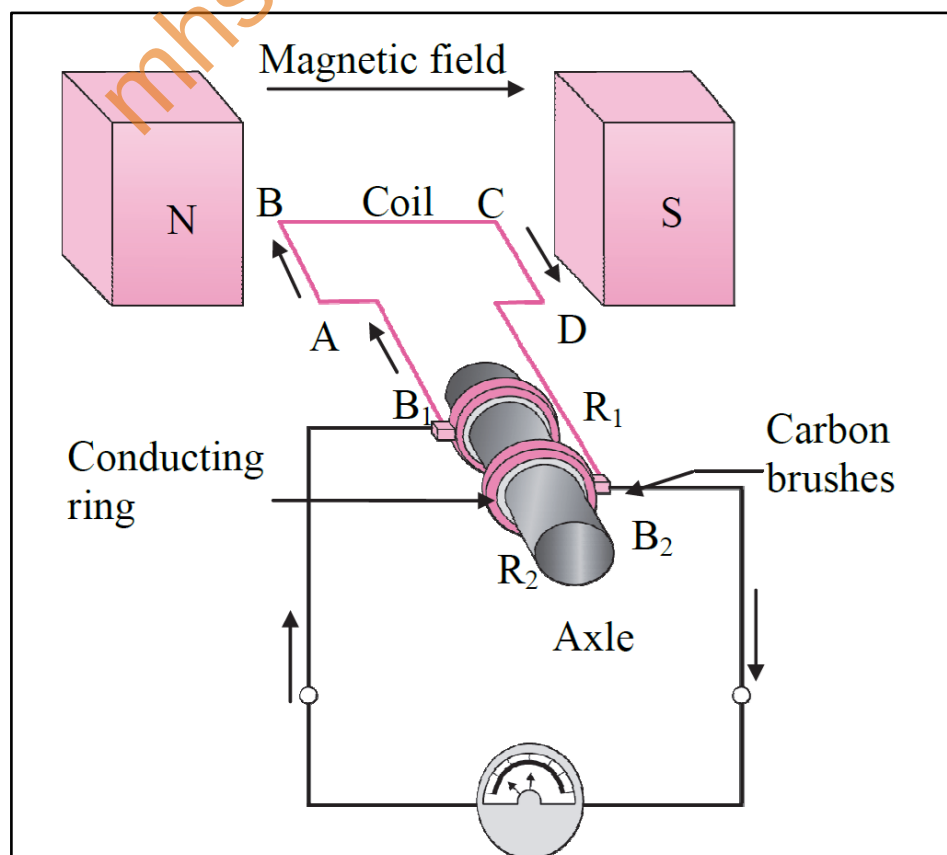
9. Who will spend more electrical energy? 500 W TV Set in 30 mins, or 600 W heater in 20 mins? – Numericals at end.
10. Observe the given figure of Fleming's Left Hand Rule and write the labels of 'A' and 'B':



A – Direction of magnetic field

B – Direction of current

11. Observe the following diagram and answer the questions given below:



a. Identify the above diagram.

Electric Generator (AC)

b. Write the principle on which the above appliance works.

Fleming's Right-hand rule (**Exercise Question 10 – a**)

c. Write the working of the above appliance.

Exercise Question 2 – b

d. Write the use of the above appliance.

To generate AC electrical energy

12. For electric power transmission, copper or aluminium wire is used.

– **Exercise Question 7 - c**

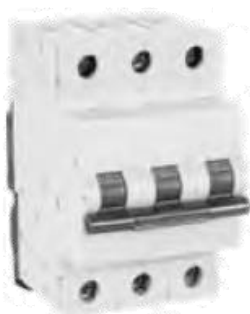
13. Identify figures A, B, C and given their uses:

(A)



Exercise Question 11 - a

(B)



Exercise Question 11 - b

(C)



Figure (c) represents a Galvanometer.

- i. It is a sensitive device which works on the same principle as that of an electric motor.
- ii. We can make some electrical measurements with it.
- iii. A coil is positioned between the pole pieces of a magnet in such a way that the pointer on the galvanometer dial is connected to it.
- iv. When a small current (for example 1 mA) flows through the coil, the coil will rotate.
- v. The rotation will be proportional to the current.
- vi. In galvanometer, the pointer deflects on both the sides of the zero-mark depending on the direction of the current.

All IMP FORMULAE

$$\text{Electric Current (I)} \equiv \frac{\text{charge (Q)}}{\text{time (t)}} \equiv \frac{\text{Coulomb}}{\text{Second}}$$

Unit - **AMPERE.**

$$\text{P.D (V)} = \frac{\text{Work done (W)}}{\text{Charge (Q)}}$$

Unit - **VOLTS.**

OHM'S LAW

$$\mathbf{V = IR}$$

All IMP FORMULAE

POWER

$$P =$$

$$P =$$

$$P = \frac{\quad}{R}$$

Unit - Watt

HEAT = P x t

$$H =$$

$$H =$$

$$H = \frac{\quad}{R}$$

Unit - Joule

Heat Produced = Energy consumed

